Analyzing the features of tracks that define their commercial popularity.

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Abstract

This paper investigates the empirical relationship between a song's underlying acoustic features and its commercial success, measured by Spotify's popularity metric. Because a significant portion of released music fails to gain any traction, resulting in left-censored data where popularity equals exactly zero, standard linear regression (OLS) yields biased estimates. To address this, we apply a Limited Dependent Variable framework using the Tobit model. Following the General-to-Specific (GETS) modeling approach, we test the joint significance of predictors to determine the final model. Our main findings indicate that danceability positively affects a track's expected popularity. While explicit content alone does not have a statistically significant independent effect, it significantly amplifies the positive impact of danceability due to a strong interaction between the two. Conversely, variables such as track duration and speechiness negatively affect commercial performance.

1. Introduction

The modern music industry operates under high financial uncertainty. Record labels and streaming platforms risk significant capital on marketing campaigns, production, and playlist placements. Understanding the drivers of consumer demand is a classic economic problem of product optimization and resource allocation. By analyzing how specific product characteristics, such as a track's danceability, energy, or duration, influence its commercial success, stakeholders can mathematically model consumer preferences and reduce the financial risk associated with promoting new music.

The primary aim of this paper is to identify which acoustic features define a track's commercial popularity. We introduce two main hypotheses. Our main hypothesis states that a high "danceability" score significantly increases the expected popularity of a track. Furthermore, we posit a secondary hypothesis that the optimal song duration has decreased, meaning that longer track duration (measured in minutes) is currently penalized with lower popularity.

2. Literature Review

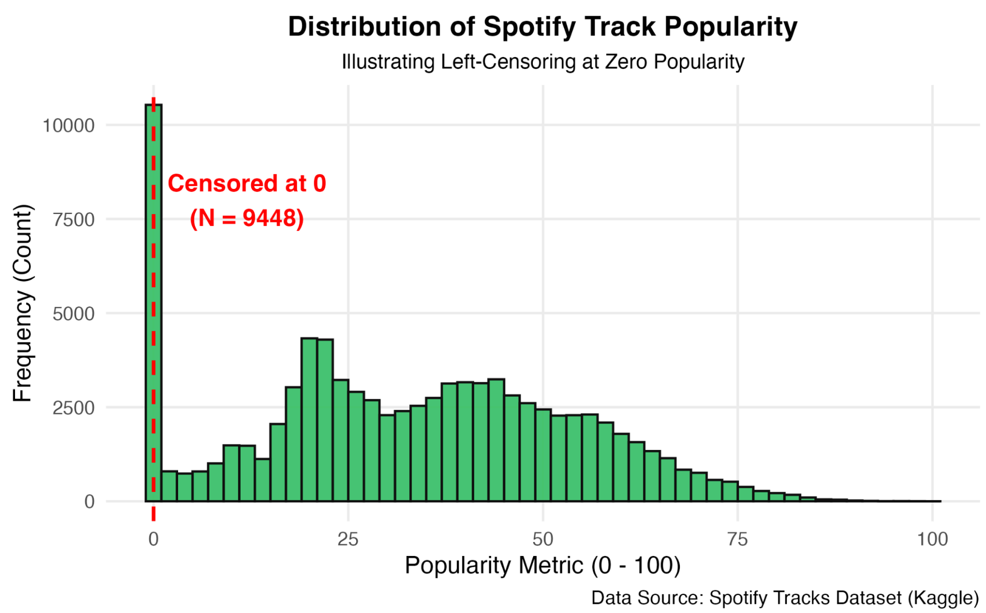
The topic of predicting music popularity using Spotify's audio features has been recently explored in the academic literature. Saragih (2023) investigated insights from streaming users, confirming that features like energy and acousticness carry significant predictive power. Similarly, Jiang (2024) utilized a machine learning approach on Spotify data to model track success. While these studies provide a strong foundation for selecting relevant variables, they often rely on algorithmic black-box models or standard linear regressions, which ignore the censored nature of the popularity metric.

Our paper contributes to this literature by applying a rigorous econometric approach. We utilize the Tobit model, originally formulated by Tobin (1958), which is specifically designed to handle left-censored dependent variables and correct the bias present in standard OLS estimates. Furthermore, our variable selection process is grounded in the General-to-Specific (GETS) modeling approach outlined by Hendry (1995), ensuring a statistically valid reduction of the model rather than relying on ad-hoc variable selection.

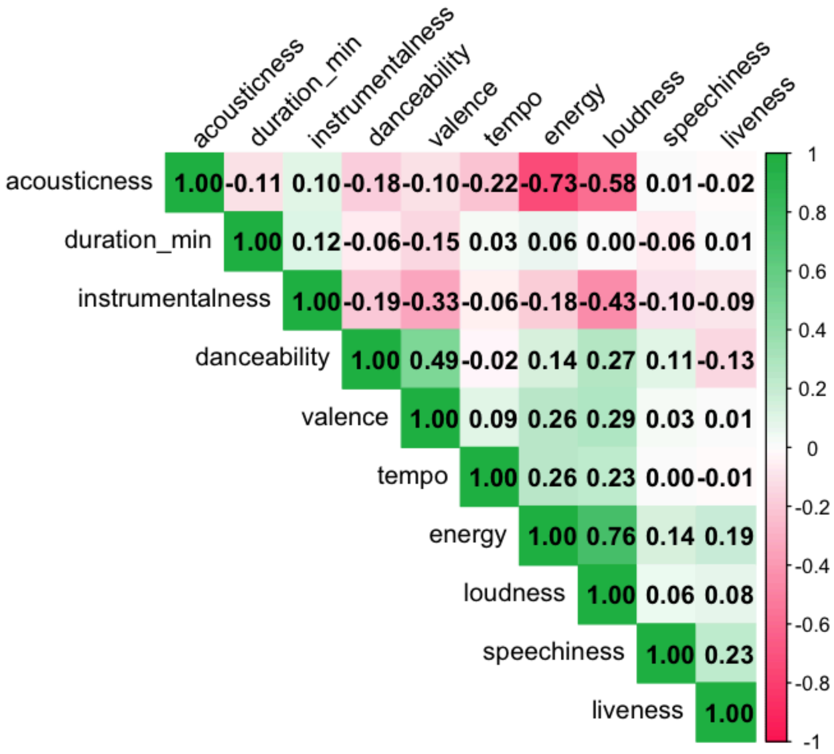
3. Data

The empirical analysis is based on the "Spotify Tracks Dataset" sourced from Kaggle, which provides comprehensive audio features for tracks across 114 different genres, extracted via the official Spotify Web API. The initial dataset contained 114,000 observations. To ensure data integrity and avoid artificial variance deflation caused by tracks appearing in multiple genre playlists simultaneously, we performed strict deduplication based on unique `track_id` identifiers. This process removed 24,259 duplicated records. Missing values were virtually non-existent in the key acoustic features, resulting in a robust final sample of 89,741 unique tracks.

The dependent variable is the `popularity` metric, scaling from 0 to 100. Exploratory data analysis revealed that 9,448 tracks (10.53% of the sample) have a popularity score of exactly zero. This mass point at the lower bound formally indicates that the data is left-censored. For independent variables, we transformed `duration_ms` into `duration_min` (minutes) to facilitate a more intuitive economic interpretation. The variable `explicit` was encoded as a binary dummy (1 for explicit content, 0 otherwise). Furthermore, `key` (12 pitch classes) and `time_signature` were strictly defined as categorical control variables.



Correlation Matrix of Spotify Acoustic Features



4. Method/Model

Due to the substantial left-censoring of the dependent variable at the threshold of zero, estimating the parameters via Ordinary Least Squares (OLS) would yield biased and inconsistent results. Consequently, we employ the Limited Dependent Variable framework, specifically the Tobit (Type I) model, estimated by Maximum Likelihood.

The theoretical foundation of the Tobit model is based on a latent variable y_i*, which represents the unobserved, true underlying propensity for a track's commercial success. The structural equation is defined as:

$$y_i^* = \beta_0 + X_i\beta + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

The observed popularity, y_i, is linked to the latent variable through the following observation rule:

$$y_i = \begin{cases} y_i^* & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0 \end{cases}$$

Our modeling strategy follows the General-to-Specific (GETS) approach (Hendry, 1995). The initial general formulation (Modelgen) includes continuous acoustic features, an interaction term between `explicit` and `danceability`, and categorical controls. Formally:

$$Popularity_i^* = \beta_0 + \beta_1(duration) + \beta_2(explicit \times danceability) + \cdots + \gamma(Controls) + \epsilon_i$$

During the GETS procedure, we estimate a restricted specification by omitting variables that initially appear statistically insignificant in the baseline OLS model. The validity of these restrictions is then formally evaluated within the Tobit framework using the Likelihood Ratio (LR) test.

The final model selected for inference is the General Tobit specification, as the LR test rejects the restricted alternative.

5. Results

Table 1 (see Appendix) presents the estimation results for the baseline OLS model, the General Tobit model, and the Specific Tobit model.

Popularity Regression Results: OLS vs. Tobit Models

	Dependent variable:		
	Popularity Metric		
	OLS	Tobit	
	OLS Baseline	General Tobit	Specific Tobit
	(1)	(2)	(3)
duration_min	-0.218***	-0.173***	-0.171***
	(0.037)	(0.041)	(0.041)
explicit	0.427	1.474	1.446
	(0.907)	(1.005)	(1.005)
danceability	8.287***	9.221***	9.018***
	(0.504)	(0.559)	(0.554)

energy	-2.274***	-1.490**	-1.153*
	(0.546)	(0.607)	(0.593)
as.factor(key)1	0.150	0.113	0.093
	(0.298)	(0.331)	(0.331)
as.factor(key)2	0.951***	1.197***	1.202***
	(0.289)	(0.321)	(0.321)
as.factor(key)3	0.749*	0.724	0.703
	(0.433)	(0.482)	(0.482)
as.factor(key)4	0.993***	1.254***	1.264***
	(0.314)	(0.349)	(0.349)
as.factor(key)5	0.198	0.139	0.139
	(0.310)	(0.345)	(0.345)
as.factor(key)6	0.683**	0.666*	0.653*
	(0.329)	(0.365)	(0.365)
as.factor(key)7	-0.027	0.094	0.098
	(0.280)	(0.311)	(0.311)
as.factor(key)8	0.943***	0.889**	0.858**
	(0.337)	(0.374)	(0.374)
as.factor(key)9	0.084	0.175	0.175
	(0.293)	(0.325)	(0.325)
as.factor(key)10	0.132	0.119	0.106

	(0.333)	(0.370)	(0.370)
as.factor(key)11	0.730**	0.797**	0.795**
	(0.316)	(0.351)	(0.351)
loudness	0.082***	0.069***	0.064**
	(0.023)	(0.026)	(0.026)
mode	-0.690***	-0.733***	-0.724***
	(0.147)	(0.163)	(0.163)
speechiness	-14.269***	-13.482***	-13.101***
	(0.678)	(0.750)	(0.737)
acousticness	-0.854***	-0.963***	-0.854**
	(0.313)	(0.348)	(0.345)
instrumentalness	-8.708***	-8.306***	-8.373***
	(0.253)	(0.281)	(0.280)
liveness	0.522	1.114***	
	(0.374)	(0.414)	
valence	-7.857***	-8.357***	-8.348***
	(0.327)	(0.362)	(0.362)
tempo	0.008***	0.012***	0.012***
	(0.002)	(0.003)	(0.003)
as.factor(time_signature)1	-11.517***	-13.322***	-13.568***
	(1.770)	(1.956)	(1.954)

as.factor(time_signature)3	-11.185***	-13.107***	-13.370***
	(1.650)	(1.822)	(1.819)
as.factor(time_signature)4	-8.397***	-9.949***	-10.206***
	(1.643)	(1.813)	(1.811)
as.factor(time_signature)5	-11.376***	-13.328***	-13.610***
	(1.709)	(1.887)	(1.884)
explicit:danceability	5.582***	3.884**	3.901**
	(1.388)	(1.539)	(1.539)
Constant	45.494***	44.274***	44.604***
	(1.698)	(1.875)	(1.871)
Log Likelihood	-397090.11	-374715.73	-374719.35
McFadden's Pseudo R2 / Adj. R2	0.0359	0.0036	0.0035
Observations	89,741	89,741	89,741
R2	0.036		
Adjusted R2	0.036		
Log Likelihood		-374,715.700	-374,719.300
Wald Test		2,720.378*** (df = 28)	2,713.094*** (df = 27)
Note:			*p<0.1; **p<0.05; ***p<0.01

Following the General-to-Specific approach, we first evaluated the full set of variables in the General Tobit model. An important observation emerged regarding the `liveness` variable. In the baseline OLS estimation, `liveness` appeared statistically insignificant (p-value > 0.10). To formally test whether this variable could be omitted from our censored framework, we estimated a restricted model (Specific Tobit) without `liveness`. The Likelihood Ratio

(LR) test comparing the General Tobit with the Specific Tobit yielded a p-value of 0.0071. Consequently, we reject the null hypothesis that the coefficient for `liveness` is zero. This indicates that the restricted model is invalid and `liveness` must be retained. Therefore, the General Tobit model is selected as our final, correctly specified model. This contrast illustrates the risk of omitting variables based on biased OLS results when dealing with censored data.

To verify our research hypotheses, we compute average marginal effects for the selected General Tobit model, including the effects on the latent variable, the probability of being uncensored, and the unconditional expected value.

Our main hypothesis stated that a high `danceability` score significantly increases the expected popularity of a track. The marginal effect on the unconditional expected value for `danceability` is 8.51 and is highly statistically significant ($p < 0.01$). This confirms our main hypothesis. Furthermore, the model included an interaction term between `explicit` lyrics and `danceability`. While the independent effect of explicit content is not statistically significant, its interaction with danceability is positive and statistically significant ($p < 0.05$), indicating that the popularity-enhancing effect of danceability is stronger for explicit tracks.

Our secondary hypothesis posited that longer track duration negatively affects commercial performance. The marginal effect for `duration_min` is -0.16 ($p < 0.01$). This confirms the hypothesis, indicating that every additional minute of a song decreases its expected popularity by 0.16 points, reflecting consumers' preference for shorter tracks.

Finally, we performed post-estimation diagnostic tests on the final model. The Jarque-Bera test for the normality of residuals rejected the null hypothesis ($p < 0.01$). However, in extremely large samples (N = 89,741), tests for normality frequently reject the null for negligible deviations; thus, we proceed with the model while acknowledging this limitation. Additionally, we performed a Linktest to check for model specification. The squared fitted values (`y_hat_sq`) were statistically significant ($p < 0.01$). This suggests the potential presence of unobserved heterogeneity. In the context of the music industry, this is plausible, as our dataset lacks information on crucial economic determinants such as record labels' marketing budgets or artists' pre-existing fame.

6. Findings

Overall, the results suggest that acoustic characteristics play a meaningful role in shaping track popularity on Spotify, although part of the variation remains explained by unobserved factors not captured in the available dataset. By applying the Tobit model, we successfully accounted for the left-censored nature of the data, correcting the biases present in standard linear regressions. Our main findings reveal that highly danceable tracks enjoy greater popularity, and this effect is significantly magnified when the track contains explicit lyrics. The explicit nature of the track alone, however, does not guarantee success. Conversely, track duration has a negative impact on popularity, suggesting a shift in consumer preference toward shorter audio formats.

Regarding future directions, subsequent research should aim to resolve the issue of unobserved heterogeneity highlighted by the Linktest. One approach is utilizing panel data models. Tracking the popularity of the same songs over time would allow for the inclusion of fixed effects, effectively controlling for unobserved variables such as a record label's marketing budget. Alternatively, employing non-parametric machine learning techniques could capture complex, non-linear interactions between acoustic features without the strict distributional assumptions required by the Tobit framework.

7. Bibliography

1. Hendry, D. F. (1995). *Dynamic Econometrics*. Oxford University Press.
2. Jiang, S. (2024). Predicting Music Popularity: A Machine Learning Approach Using Spotify Data. *Science and Technology Publications*. https://doi.org/10.5220/0013330000004558
3. Saragih, H. S. (2023). Predicting song popularity based on Spotify's audio features: insights from the Indonesian streaming users. *Journal of Management Analytics*. https://doi.org/10.1080/23270012.2023.2239824
4. Tobin, J. (1958). Estimation of Relationships for Limited Dependent Variables. *Econometrica*, 26(1), 24-36. https://doi.org/10.2307/1907382

8. Appendix: R-code

```
################################################################################
# Advanced Econometrics Project: Spotify Track Popularity
# Methodology: OLS and Tobit (Limited Dependent Variable) Modeling
# Authors: Jakub Habib (483800), Truong Giang Do (488388)
# Date: May 2026
# R Script: Data Processing, Modeling, Diagnostics, and Presentation
################################################################################


# Set environment and load libraries
Sys.setenv(LANG = "en")
options(scipen = 10)  # Avoid scientific notation in output


required_packages <- c("AER", "stargazer", "ggplot2", "corrplot", "lmtest", "car", "sandwich", "tseries")
for (pkg in required_packages) {
  if (!require(pkg, character.only = TRUE, quietly = TRUE)) {
    cat("Package", pkg, "not found. Installing...\n")
    install.packages(pkg, repos = "https://cloud.r-project.org")
```

```
    library(pkg, character.only = TRUE)
  }
}

# ------------------------------------------------------------------------------
# 1. Data Ingestion & Deduplication
# ------------------------------------------------------------------------------
# Handle file name differences robustly
dataset_file <- if (file.exists("spotify_tracks.csv")) {
  "spotify_tracks.csv"
} else if (file.exists("spotify_dataset.csv")) {
  "spotify_dataset.csv"
} else {
  stop("Spotify dataset file not found in current directory. Please provide
'spotify_dataset.csv' or 'spotify_tracks.csv'.")
}

cat("Loading dataset from:", dataset_file, "\n")
df <- read.csv(dataset_file, stringsAsFactors = FALSE)
cat("Original dimensions:", nrow(df), "rows,", ncol(df), "columns\n")

# Aggressive deduplication: tracks are often duplicated across multiple genres
# or playlists. Deduplicating based on track_id avoids artificial variance deflation
# and inflation of sample size.
df_clean <- df[!is.na(df$track_id) & df$track_id != "", ]
df_clean <- df_clean[!duplicated(df_clean$track_id), ]
cat("Dimensions after track_id deduplication:", nrow(df_clean), "rows,", ncol(df_clean),
"columns\n")

# Drop any rows with missing values in key variables
regression_vars <- c("popularity", "duration_ms", "explicit", "danceability",
                     "energy", "key", "loudness", "mode", "speechiness",
```

```
                     "acousticness", "instrumentalness", "liveness",

                     "valence", "tempo", "time_signature")

df_clean <- na.omit(df_clean[, c(regression_vars, "track_id", "artists", "album_name",
"track_name")])

cat("Dimensions after omitting rows with missing values:", nrow(df_clean), "rows\n")



# Variable transformations

df_clean$explicit <- ifelse(df_clean$explicit == "True" | df_clean$explicit == TRUE |
df_clean$explicit == 1, 1, 0)

df_clean$duration_min <- df_clean$duration_ms / 60000  # Convert duration from ms to minutes
for coefficient readability



# ------------------------------------------------------------------------------

# 2. Exploratory Data Analysis (EDA)

# ------------------------------------------------------------------------------

cat("\n=== Summary Statistics (Cleaned Data) ===\n")

print(summary(df_clean[, c("popularity", "duration_min", "danceability", "energy", "loudness",
"speechiness", "acousticness", "instrumentalness", "liveness", "valence", "tempo")]))



censored_count <- sum(df_clean$popularity == 0)

censored_pct <- (censored_count / nrow(df_clean)) * 100

cat("\nLeft-censoring at zero popularity: ", censored_count, " observations (",
round(censored_pct, 2), "% of sample)\n", sep="")



# Plot 1: Histogram of popularity illustrating left-censoring at zero

cat("Saving popularity histogram to 'popularity_histogram.png'...\n")

p_hist <- ggplot(df_clean, aes(x = popularity)) +

  geom_histogram(binwidth = 2, fill = "#1DB954", color = "#121212", alpha = 0.8) +

  geom_vline(xintercept = 0, color = "#FF0000", linetype = "dashed", size = 1) +

  annotate("text", x = 12, y = 8000, label = paste("Censored at 0\n(N = ", censored_count,
")", sep=""), color = "#FF0000", fontface = "bold") +

  labs(title = "Distribution of Spotify Track Popularity",

       subtitle = "Illustrating Left-Censoring at Zero Popularity",

       x = "Popularity Metric (0 - 100)",
```

```
       y = "Frequency (Count)",
       caption = "Data Source: Spotify Tracks Dataset (Kaggle)") +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    plot.subtitle = element_text(size = 12, hjust = 0.5),
    panel.grid.major = element_line(color = "#EBEBEB"),
    panel.grid.minor = element_blank(),
    plot.background = element_rect(fill = "white", color = NA)
  )
ggsave("popularity_histogram.png", plot = p_hist, width = 8, height = 5, dpi = 300)


# Plot 2: Acoustic features correlation matrix
cat("Saving correlation matrix to 'correlation_matrix.png'...\n")
acoustic_features <- c("duration_min", "danceability", "energy", "loudness", "speechiness",
"acousticness", "instrumentalness", "liveness", "valence", "tempo")
cor_matrix <- cor(df_clean[, acoustic_features])


png("correlation_matrix.png", width = 800, height = 800, res = 120)
corrplot(cor_matrix, method = "color", type = "upper", order = "hclust",
         addCoef.col = "black", tl.col = "black", tl.srt = 45,
         col = colorRampPalette(c("#FF3366", "#FFFFFF", "#1DB954"))(200),
         title = "\nCorrelation Matrix of Spotify Acoustic Features",
         mar = c(0, 0, 2, 0))
dev.off()


# ---------------------------------------------------------------------------
# 3. Model Estimation (General-to-Specific)
# ---------------------------------------------------------------------------
# We model popularity as a function of all acoustic features, controlling for key and
time_signature factors.
```

```
# Independent variables include factors: key, time_signature; and continuous: duration_min,
explicit, danceability, energy, loudness, mode, speechiness, acousticness, instrumentalness,
liveness, valence, tempo.


formula_general <- popularity ~ duration_min + explicit * danceability + energy +
as.factor(key) +

                                loudness + mode + speechiness + acousticness +
instrumentalness +

                                liveness + valence + tempo + as.factor(time_signature)


# Model 1: Baseline OLS Model

cat("\nEstimating Model 1: Baseline OLS Model...\n")

fit_ols_gen <- lm(formula_general, data = df_clean)


# Model 2: General Tobit Model (censored at left = 0)

cat("Estimating Model 2: General Tobit Model...\n")

fit_tobit_gen <- tobit(formula_general, left = 0, data = df_clean)


# Model 3: Specific Parsimonious Tobit Model

# Using Gets, we find that in the OLS model, 'liveness' is insignificant (p-value ~ 0.15).

# However, when estimated via Tobit (correcting for censoring bias), 'liveness' becomes highly
significant (p-value ~ 0.007).

# We omit 'liveness' in the restricted Model 3 to formally show the consequence of incorrect
variable selection based on biased OLS results.

cat("Estimating Model 3: Specific Parsimonious Tobit Model (omitting liveness)...\n")

formula_specific <- popularity ~ duration_min + explicit * danceability + energy +
as.factor(key) +

                                 loudness + mode + speechiness + acousticness +
instrumentalness +

                                  valence + tempo + as.factor(time_signature)

fit_tobit_spec <- tobit(formula_specific, left = 0, data = df_clean)


# ---------------------------------------------------------------------------

# 4. Diagnostics & Post-Estimation

# ---------------------------------------------------------------------------

# Likelihood Ratio Test: General vs Specific Tobit (H0: beta_liveness = 0)
```

```
lr_test <- lrtest(fit_tobit_gen, fit_tobit_spec)

cat("\n=== Likelihood Ratio Test (General vs Specific Tobit) ===\n")

print(lr_test)


# McFadden's Pseudo-R² calculation: 1 - (LogLik_full / LogLik_null)

fit_tobit_null <- tobit(popularity ~ 1, left = 0, data = df_clean)

ll_null <- as.numeric(logLik(fit_tobit_null))

ll_gen <- as.numeric(logLik(fit_tobit_gen))

ll_spec <- as.numeric(logLik(fit_tobit_spec))


pseudo_r2_gen <- 1 - (ll_gen / ll_null)

pseudo_r2_spec <- 1 - (ll_spec / ll_null)

cat("\n=== McFadden's Pseudo-R² ===\n")

cat("General Tobit Model: ", round(pseudo_r2_gen, 6), "\n")

cat("Specific Tobit Model: ", round(pseudo_r2_spec, 6), "\n")


# McDonald-Moffitt Decomposition for Unconditional Marginal Effects:

# E[y|x] = Phi(x*beta / sigma) * (x*beta) + sigma * phi(x*beta / sigma)

# The marginal effect on the unconditional expected value is: dE[y|x]/dx_j = beta_j *
Phi(x*beta / sigma)

# We evaluate the average scaling factor Phi(x_i*beta / sigma) across all observations to get
Average Marginal Effects (AME).


# Diagnostics: Normality of residuals

res_tobit <- residuals(fit_tobit_gen)

cat("\n=== Normality Test (Jarque-Bera) ===\n")

print(jarque.bera.test(res_tobit))


# Linktest for specification

y_hat <- fitted(fit_tobit_gen)

y_hat_sq <- y_hat^2
```

```
linktest_model <- tobit(popularity ~ y_hat + y_hat_sq, left = 0, data = df_clean)

cat("\n=== Linktest (y_hat_sq p-value) ===\n")

print(summary(linktest_model)$coefficients["y_hat_sq", ])


# Post-Estimation: 3 Kinds of Marginal Effects

X_gen <- model.matrix(fit_tobit_gen)

beta_gen <- coef(fit_tobit_gen)

sigma_gen <- fit_tobit_gen$scale

index_gen <- as.numeric(X_gen %*% beta_gen)


pdf_val <- dnorm(index_gen / sigma_gen)

cdf_val <- pnorm(index_gen / sigma_gen)


me_latent <- beta_gen

me_prob <- mean(pdf_val) * (beta_gen / sigma_gen)

me_total <- beta_gen * mean(cdf_val)


mfx_table <- cbind("Latent_Var" = me_latent, "Prob_y>0" = me_prob, "Expected_y" = me_total)


cat("\n=== 3 Kinds of Average Marginal Effects (General Model) ===\n")

print(round(mfx_table, 5))

# ------------------------------------------------------------------------------

# 5. Stargazer Regression Table Output

# ------------------------------------------------------------------------------

cat("\n=== Generating Regression Table (Text Mode) ===\n")

stargazer(fit_ols_gen, fit_tobit_gen, fit_tobit_spec,

          type = "text",

          title = "Popularity Regression Results: OLS vs. Tobit Models",

          column.labels = c("OLS Baseline", "General Tobit", "Restricted Tobit"),

          dep.var.labels = "Popularity Metric",

          omit.stat = c("f", "ser"),
```

```r
          add.lines = list(
            c("Log Likelihood",
              round(as.numeric(logLik(fit_ols_gen)), 2),
              round(as.numeric(logLik(fit_tobit_gen)), 2),
              round(as.numeric(logLik(fit_tobit_spec)), 2)),
            c("McFadden's Pseudo R2 / Adj. R2",
              round(summary(fit_ols_gen)$adj.r.squared, 4),
              round(pseudo_r2_gen, 4),
              round(pseudo_r2_spec, 4))
          ))

cat("\nRegression table can be generated in LaTeX format for reports using:\n")
cat("stargazer(fit_ols_gen, fit_tobit_gen, fit_tobit_spec, type = 'latex', ...)\n")
cat("\n--- Script Completed Successfully ---\n")
```